

Mandarin Verb Doubling in Locative and Directional Phrases: Structures in Syntax and Postsyntax

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1. Introduction

Mandarin verb doubling (VD) comprises configurations with two phonologically identical verb occurrences. The first is realised within a verbal predicate-sized constituent, typically with the object; the second hosts aspectual or construction-specific morphology and precedes postverbal material, which may be an adjunct, secondary predicate, or argument. Not all such material is restricted to following the second verb. I argue that locative and directional spatial phrases (SpXPs) (C. N. Li and Thompson 1981, pp. 443–444) are more tightly integrated into predicate structure than frequency, duration, or manner modifiers, revealing a division of labour between syntax and postsyntax. In (1), the first verb occurs post-subject and is pronounced with phrasal material: *xie ji-ge zi* ‘write several words’ in the locative example and *san ji-xiaoshi bu* ‘go for a several-hour walk’ in the directional example. Consistent with Yan (2025), the SpXP is pronounced with the second verb. Bold brackets identify SpXP constituents throughout the examples.

- (1) a. Piet xie ji-ge zi xie-[**zai-le qiang-mian**].
Piet write several-CL word write-LOC-PFV wall-surface
‘Piet wrote several words on the wall.’
b. Piet san ji-xiaoshi bu san-[**qu-le gongyuan**].
Piet walk several-hour step walk-DIR-PFV park
‘Piet went for a several-hour walk to the park.’

Section 2 uses diagnostics to distinguish preverbal modifier SpXPs from postverbal predicate-internal SpXPs. Section 3 extends the postsyntactic operations of Meadows and Yan (2023) to verbal complexes containing spatial and aspectual morphology. Section 4 concludes with implications for multiple constituents in the VD postverbal field.

2. Locative and directional verb doubling

The adjunct-island contrast in (2) provides evidence that these SpXP VD configurations involve phrasal movement. The higher, pronounced VP copy remains in its ordinary post-subject position, while the lower verbal occurrence and its associated SpXP are embedded inside a *yinwei* ‘because’ adjunct. If the first occurrence were independently base-generated in its surface position, embedding the lower predicate inside an adjunct should not eliminate the intended VD interpretation. The paradigm adapts the adjunct-island diagnostic presented by Meadows and Yan (2023, 114, ex. 17) to locative and directional SpXP VD; Yan (2025, p. 11) reports parallel effects with other types of postverbal material.

- (2) a. *Piet xie ji-ge zi yinwei xie-[**zai-le qiang-mian**].
Piet write several-CL word because write-LOC-PFV wall-surface
Intended: ‘Piet wrote several words because he wrote them on the wall.’
b. *Piet san ji-xiaoshi bu yinwei san-[**qu-le gongyuan**].
Piet walk several-hour step because walk-DIR-PFV park
Intended: ‘Piet went for a several-hour walk because he went to the park.’

Having established a phrasal-movement dependency from adjunct island effects, this section examines SpXP placement relative to the two verbal copies. Frequency, duration, subject-oriented depictive, and resultative expressions are confined to the lower postverbal field, whereas SpXPs may follow the lower verb or occur between the copies (between-copy SpXP).¹

This flexibility raises a constituency question. Linear adjacency to the higher copy does not establish that a between-copy SpXP is contained in that phrase; the SpXP may instead be base-generated as a modifier of the lower predicate. Conversely, in canonical postverbal SpXP VD, the higher copy is pronounced without the SpXP, while the SpXP remains in the lower predicate domain. The diagnostics below exploit this distributional asymmetry to distinguish predicate-internal, argumental SpXPs selected by the lower predicate from modifier-like SpXPs in the between-copy position. I analyse this asymmetry in first-phase syntax, following Tham (2025) on Mandarin *zai*-phrases as event modifiers. I first compare other postverbal VD types, assuming that they derive from non-VD counterparts.

2.1 Parallels with other types of VD

The adjunct island parallel extends to the familiar, non-SpXP VD configurations involving frequency, duration, manner, and result-state material in the postverbal field. This parallel shows that SpXP VD belongs to the same structural family as the better-studied non-spatial VD types (Cheng 2007; Cheng 2017; Bartos 2019; Yan 2025; Lai 2021; Huang 2018). All four examples share the prefix *Piet du ji-ben shu* (Piet read several-CL book) ‘Piet read several books’; (**yinwei*) marks the adjunct-island violation.

Type	Lower continuation and gloss	Reading
Frequency	(<i>*yinwei</i>) <i>du-le san bian</i> because read-PFV three time	three times
Manner	(<i>*yinwei</i>) <i>du-de hen zixi</i> because read-DE very carefully	very carefully
Result state	(<i>*yinwei</i>) <i>du-de kaixin</i> because read-DE happy	and became happy
Duration	(<i>*yinwei</i>) <i>du-le yi xingqi</i> because read-PFV one week	for one week

A second parallel concerns overt functional material in the lower verbal domain, assuming the derivation relation between non-VD and VD pairs. Frequency and duration VD realise *-le* on the second verbal occurrence, whereas manner and result-state VD use *de*. SpXP VD patterns with the former in requiring overt aspect on the lower verbal complex (Meadows and Yan 2025, 430, exx. 10–11). The corresponding non-VD clauses identify the aspectual host: under the intended bounded episodic reading, *-le* attaches to the lexical predicate rather than to the spatial head.

- (4) a. Piet xie-le ji-ge zi [zai qiang-mian].
Piet write-PFV several-CL word LOC wall-surface
‘Piet wrote several words on the wall.’

¹The analysis below also derives SpXPs pronounced with the higher copy:

Piet xie ji-ge zi [zai qiang-shang] xie-le hao-ji-hui.

Piet write several-CL word LOC wall-top write-PFV several-times

‘Piet wrote several words on the wall several times.’

- b. Piet san-le ji-xiaoshi bu [qu gongyuan].
 Piet walk-PFV several-hour step DIR park
 ‘Piet went for a several-hour walk to the park.’

Without *-le* on *xie* or *san*, the clauses do not receive the intended bounded episodic interpretation; imperative reading may remain available. Conversely, replacing bare *zai* or *qu* in (4) with *zai-le/guo* or *qu-le/guo* is impossible.²

This restriction distinguishes SpXP VD from ordinary serial verb constructions. In serial verb configurations with two lexical verbs, aspectual marking is more flexible: perfective *-le* may appear on either verb, as illustrated in (5), adapted from Y. Li (1991).

- (5) a. Ta na dao qie-le rou.
 3SG take knife cut-PFV meat
 ‘He cut the meat with a knife.’ (Y. Li 1991, 104, ex. 11)
 b. Ta na-le dao qie rou.
 3SG take-PFV knife cut meat
 ‘He took the knife to cut meat.’ (Y. Li 1991, 112, ex. 13a)

2.2 Locating SpXPs in verbal predicates

On the basis of linear order alone, SpXP VD resembles VD constructions with frequency and duration expressions in the lower postverbal field. The resemblance is only superficial. Frequency phrases quantify the number of event occurrences, while duration phrases measure an event’s temporal extent; their licensing therefore depends on event measurement. Postverbal SpXPs, by contrast, must be integrated into the predicate’s internal structure, assuming an identity with Ramchand (2008)’s Res/Proc projection as a location or path. Their distribution is sensitive to the lexical and event-structural properties of the verbal predicate. As shown by Yan (2025, 31, ex. 48), replacing *xie ji-ge zi* ‘write several characters’ or *san ji-xiaoshi bu* ‘go for a several-hour walk’ with *du ji-ben shu* ‘read several books’ renders the corresponding postverbal locative and directional VD configurations unacceptable.

- (6) a. *Piet du ji-ben shu du-[zai-le chuang-shang].
 Piet read several-CL book read-LOC-PFV bed-on
 Intended: ‘Piet read books on the bed.’
 b. *Piet du ji-ben shu du-[qu-le tushuguan].
 Piet read several-CL book read-DIR-PFV library
 Intended: ‘Piet went to the library to read books.’

2.3 Preverbal SpXPs and the complement–modifier contrast

The contrast just discussed does not mean that predicates such as *du ji-ben shu* ‘read several books’ are incompatible with spatial material in general. The same verbal predicate can host a locative or directional SpXP in positions distinct from the postverbal field occupied by frequency and duration phrases. In particular, when the SpXP precedes the verbal predicate *du ji-ben shu*, both locative and directional configurations and their VD counterparts are acceptable.

- (7) a. Piet [zai chuang-shang] du ji-ben shu.
 Piet LOC bed-on read several-CL book
 ‘Piet read several books on the bed.’

²If *zai* is analysed as a preposition, its inability to host aspect follows independently from its categorial status. For *qu*, the restriction concerns its use as a directional head in the present configuration; lexical *qu* ‘go’ can independently bear aspectual morphology.

- b. Piet [**qu tushuguan**] du ji-ben shu.
 Piet DIR library read several-CL book
 ‘Piet went to the library to read several books.’

The examples in (7) show that there is no lexical incompatibility between *du ji-ben shu* and locative or directional material. What matters is the structural position of the SpXP. In both (7a) and (7b), the SpXP functions as an event modifier, showing designated locations for events.³

The same preverbal SpXP configuration can also occur inside a VD string. In (8), the lower predicate domain contains a preverbal SpXP followed by the second verbal occurrence. The optional temporal adverb *yiqian* makes episodic interpretation in the past accessible.

- (8) a. Piet du ji-ben shu (yiqian) [**zai chuang-shang**] du(-le).
 Piet read several-CL book formerly LOC bed-on read(-PFV)
 ‘Piet used to read several books on the bed.’
 b. Piet du ji-ben shu (yiqian) [**qu tushuguan**] du(-le).
 Piet read several-CL book formerly DIR library read(-PFV)
 ‘Piet used to go to the library to read several books.’

In (8), the SpXP precedes the second verbal occurrence. Prior to the syntax–semantics diagnostics introduced below, this configuration is structurally ambiguous. The SpXP may occupy its ordinary preverbal modifier position with respect to the second verbal form. Alternatively, the string may involve fronting of a larger predicate-sized constituent which contains the SpXP. Since these two possibilities are not distinguished by surface order alone, the examples in (8) cannot by themselves decide whether the relevant SpXP is a preverbal modifier or a predicate-internal spatial dependent. These two possibilities can be distinguished by two diagnostics: an aspectual diagnostic and an interpretive one.

To compare the two analyses, the first diagnostic tests whether an SpXP in the non-canonical between-copy position belongs to a fronted predicate-sized constituent or is base-generated in its surface position. The aspectual contrast bears on the status of this SpXP. In the postverbal pattern (9a) and (10a), the SpXP is realised as a predicate-internal spatial argument, and perfective *-le* is licensed on the lower verbal complex. In the between-copy pattern (9b) and (10b), the same spatial material precedes the second verbal occurrence. In this position, however, *-le* is degraded, whereas the bare lower verb and experiential *-guo* are acceptable.

- (9) a. Piet yiqian xie ji-ge-zi xie-[**zai-le qiangmian**].
 Piet previously write several-CL-character write-LOC-PFV wall.surface
 ‘Piet previously wrote a few characters on the wall.’ (postverbal SpXP; argumental; *-le* licensed)
 b. Piet yiqian xie ji-ge-zi [**zai qiangmian**] xie (*le)/(guo).
 Piet previously write several-CL-character LOC wall.surface write (*PFV)/(EXP)
 ‘Piet previously wrote a few characters on the wall.’ (between-copy SpXP; bare/*-guo* acceptable, *-le* degraded)
 (10) a. Piet yiqian san-bu san-[**qu-le gongyuan**].
 Piet previously walk.steps walk-DIR-PFV park
 ‘Piet previously went for a walk to the park.’ (postverbal SpXP; argumental; *-le* licensed)

³Preverbal SpXPs can stack, whereas postverbal SpXPs cannot: *Piet [qu tushuguan] [jin kafeiguan] du ji-ben shu*; Piet DIR library DIR café read several-CL book; ‘Piet went to the library to enter the café to read several books.’

- b. Piet yiqian san-bu [**qu gongyuan**] san (*le)/(guo).
 Piet previously walk.steps DIR park walk (*PFV)/(EXP)
 ‘Piet previously went to the park for a walk.’ (between-copy SpXP; bare/-*guo* acceptable, -*le* degraded)
- (11) a. Piet xie ji-ge-zi (zai zuotian) [**zai qiangmian**] (zai zuotian)
 Piet write several-CL-character (LOC yesterday) LOC wall.surface (LOC yesterday)
 xie-le.
 write-PFV
 ‘Piet wrote a few characters on the wall yesterday.’
- b. Piet san-bu (zai zuotian) [**qu gongyuan**] (zai zuotian) san-le.
 Piet walk.steps (LOC yesterday) DIR park (LOC yesterday) walk-PFV
 ‘Piet went to the park for a walk yesterday.’

The contrast is specific to the interaction between perfective *-le* and the event structure of the lower predicate. Verbal *-le* presents the event as bounded or realised, and its felicity is sensitive to the availability of an appropriate transition or event boundary (Soh and Gao 2006). Experiential *-guo*, by contrast, does not assert culmination or result attainment; it asserts that the event type has at least one prior occurrence (Iljic 2009).

The improvement produced by placing an independent temporal adjunct on either side of the between-copy SpXP shows that linear adjacency alone fails to establish constituency. In (11), the adjunct can separate the SpXP from the higher V–object copy while preserving the intended reading, supporting modifier-like attachment within the lower predicate domain.

The contrast follows from relating SpXPs to participants in a decomposed verbal structure. I adopt a first-phase syntax in which the verbal predicate is decomposed into InitP, ProcP, and ResP (Ramchand 2008). InitP introduces an initiation or causing subevent, ProcP introduces a process subevent, and ResP introduces a result-state subevent. Their specifiers are interpreted as the initiator, undergoer, and resultee, respectively. Rhematic material, by contrast, occupies a complement position and does not introduce an independent subevent; it contributes to the description of the subevent introduced by the relevant head. This distinction is crucial for SpXPs. Locative and directional SpXPs in the postverbal field are selected spatial dependents whose interpretation is determined by the subevental projection with which they combine.

The relevant distinction is between a stative result-location relation and a dynamic path relation. In the locative/result-state configuration, the SpXP is associated with ResP. It describes the ground with respect to which the resultee’s result state is evaluated. In the directional configuration, the SpXP is associated with ProcP. It contributes a path or directional measure to the process, and the moving participant is interpreted as an undergoer whose motion is mapped onto that path. The dynamicity of the directional construal therefore comes from ProcP, not from the SpXP independently. Conversely, the stative result-location construal comes from ResP, where the SpXP supplies the ground of the result state.

The event-structural contrast is isolated by holding the predicate *zou shanlu* ‘walk along the mountain road’ constant and varying only the spatial element. With *zai/dao* ‘LOC/reach’, *zui waice* ‘LOC most outer.side’ denotes Piet’s attained result location, contradicting a continuation that places other individuals farther outside. With *xiang/qu* ‘toward/DIR’, it can instead specify an oriented path without entailing endpoint attainment, so the continuation remains compatible.

- (12) a. #Piet zou shanlu zou-[**zai/dao-le** **zui waice**], danshi Piet de
 Piet walk mountain.road walk-LOC/reach-PFV most outer.side but Piet GEN
 waice xianzai hai-you qita-ren.
 outside now still-have other-person

‘Piet walked along the mountain road and ended up at/reached the outermost position, but there are still other people farther outside than Piet.’

- b. Piet zou shanlu zou-[xiang/qu-le zui waice], danshi Piet de
 Piet walk mountain.road walk-toward/DIR-PFV most outer.side but Piet GEN
 waice xianzai hai-you qita-ren.
 outside now still-have other-person

‘Piet walked along the mountain road toward the outermost position, but there are still other people farther outside than Piet.’

The contrast distinguishes a rhematic place/ground in the ResP domain from a rhematic path in the ProcP domain. The former fixes the resultee’s attained location; the latter orders the undergoer’s motion along a path without necessarily locating it at the endpoint.⁴

In first-phase syntax, the question is whether an SpXP participates in event and argument structure or functions as an event-external modifier (Tham 2025). Unlike frequency and duration phrases, selected SpXPs can alter event structure.

I assume that verbal predicates are decomposed into subevental projections. InitP introduces an initiation subevent whose specifier is interpreted as the initiator; ProcP introduces a process subevent whose specifier is interpreted as the undergoer; and ResP introduces a result-state subevent whose specifier is interpreted as the resultee. Within this system, SpXPs do not themselves introduce Init, Proc, or Res. Rather, they are rhematic or selected spatial material whose interpretation depends on the subevental head that licenses them. A directional SpXP is licensed in the process domain: it contributes a path or directional measure, and the undergoer’s motion is mapped onto that path. A locative/result-state SpXP is licensed in the result domain: it contributes the place or ground with respect to which the resultee’s final state is described.

The two configurations can be represented as in (13).

- (13) a. Locative/result-state SpXP:

$$[\text{ProcP } DP_{\text{UNDERGOER}} [\text{Proc}' \text{Proc} [\text{ResP } DP_{\text{RESULTEE}} [\text{Res}' \text{Res } SpXP_{\text{PLACE/GROUND}}]]]]$$

- b. Directional/path SpXP:

$$[\text{ProcP } DP_{\text{UNDERGOER}} [\text{Proc}' \text{Proc } SpXP_{\text{PATH/RHEME}}]]$$

The schemas in (13) capture the complement–modifier asymmetry developed in this section. Preverbal SpXPs can be merged as event-level or internal modifiers, and their interpretation is resolved pragmatically with respect to the event. Postverbal SpXPs in locative and directional VD are different. They occupy predicate-internal positions and are licensed only when the predicate provides the relevant subevental structure. A predicate that supports a result-state relation can license a locative/result-state SpXP; a predicate that supports a process with a path structure can license a directional/path SpXP.

This view also clarifies the status of *du ji-ben shu* ‘read several books’ cases discussed above. When an SpXP appears before the second verbal occurrence, it may be parsed as a modifier of the lower predicate, or as part of a larger fronted predicate. Such examples therefore do not diagnose the postverbal SpXP-VD structure directly. The diagnostic configuration is the one in which the SpXP appears in the postverbal field after the second verbal occurrence. In that position, the SpXP must be licensed by the predicate-internal Proc/Res structure.

In the next section, I present the derivational route for SpXP as complements of verbal predicates, together with the syntax–prosody mapping that motivates the PF algorithm below.

⁴The non-culminating interpretation is clearest with *xiang*. For speakers who assign *qu* an attained-goal interpretation in this environment, *qu* may pattern with the result-location variants.

3. Deriving the possible SpXP strings in VD

This section asks whether the complement–modifier contrast established for SpXPs is preserved when the same verbal predicate is realised more than once. I argue that it is. The syntax builds a phrasal movement dependency: a predicate-sized constituent is displaced from the lower verbal domain, while the lower copy remains structurally present at PF. The spell-out of the resulting chain is then regulated by postsyntactic operations. This approach requires a model in which copy pronunciation is not limited to the wholesale pronunciation of a single phrasal copy. The relevant comparison comes from predicate clefting in Modern Hebrew, where Landau (2006) argues that a VP-movement dependency can give rise to the pronunciation of both a higher predicate copy and a lower verbal copy. In (14), the infinitival predicate appears in the left periphery, while the finite verbal form is also realised in the lower clause.

- (14) Liknot et ha-prax-im, hi kan-ta.
 buy.INF ACC DEF-flower-PL she buy-PST.3F.SG
 ‘As for buying flowers, she bought them.’ (Landau 2006, ex. 8a)

I implement this idea within a late-insertion architecture of the kind assumed in Distributed Morphology (DM) (Bobaljik 2017). Narrow syntax is responsible for the movement dependency and for the structural distinction between selected SpXPs and modifier-like material. PF determines how the copies created by syntax are pronounced. The same syntactic chain can therefore receive different surface realisations depending on the relative ordering of PF operations.

The relevant PF computation has two kinds of sensitivity as proposed by Meadows and Yan (2023, p. 110). First, some operations apply to hierarchical structure before Vocabulary Insertion and linearisation. These operations can refer to copy-chain relations and to the structural positions of heads and their dependents. Second, other morphological operations apply after linearisation, when adjacency and string order become visible. This distinction is independently required for Mandarin VD: Chain Reduction regulates the pronunciation of copies, whereas Local Dislocation (Embick and Noyer 2001) sensitive to linear adjacency between the lower verbal element and neighbouring functional material.

The analysis to be developed below therefore extends Landau’s Chain Reduction approach in two respects. First, it applies the logic of multiple copy realisation to low Mandarin VD, where the pronounced higher copy remains below the subject rather than in the clausal left periphery. Second, it refines Chain Reduction into an ordered PF system that permits partial copy realisation. This refinement is necessary for SpXP VD because the surface string separates the pronounced material of the higher copy from the spatial material of the lower copy, while still preserving the complement–modifier asymmetry diagnosed in the preceding section.

3.1 Prosodic evidence

The prosodic evidence distinguishes three copy-realisation patterns. Under prosodic binarity, a prosodic phrase is penalised when it contains fewer or more than the optimal number of prosodic-word-sized units (Ishihara and Kalivoda 2022). In the postverbal SpXP pattern, the lower copy is realised as *V–Sp–le Ground*. Because affixal *-le* does not project an independent prosodic word, the relevant binary pairing is between the complex verbal word *V–Sp–le* and the Ground; the lower domain is therefore prosodically complete. In the between-copy SpXP pattern⁵ with lower *-le*, by contrast, the SpXP is pronounced outside the lower verbal complex and cannot support the

postverbal SpXP	[V–Obj ⟨SpXP _{Arg} ⟩] _{high}	...	[V–Sp–le Ground] _{low}
preverbal SpXP with lower <i>-le</i>	[V–Obj SpXP] _{high}	...	[V–le] _{low}
preverbal SpXP without <i>-le</i>	[V–Obj] _{high}	...	[SpXP _{Mod} V] _{low}

lower prosodic domain. The lower copy is then only *V-le*. Since dependent functional material such as *-le* cannot freely supply an additional prosodic word without incurring a MATCH- ω cost (Elfner 2012, p. 244), the lower domain is a singleton and violates binarity. When lower *-le* is absent, however, the SpXP can be parsed as a genuine preverbal modifier and phrased with the bare lower verb.

3.2 Ordering PF operations

The directional SpXP pattern in (15) raises a PF-ordering problem. The surface form contains two occurrences of the verbal predicate: the first occurrence appears in the post-subject position as part of the fronted predicate-sized copy, while the second occurrence appears in the lower verbal domain together with the directional SpXP and the perfective marker *-le*.

- (15) Piet san-bu san-[qu-le gongyuan].
 Piet walk.steps walk-DIR-PFV park
 ‘Piet went for a walk to the park.’

The analysis extends Landau’s Chain Reduction system for multiple copy realisation (Landau 2006); the general Chain Reduction mechanism follows Nunes (2004). I assume the two PF principles in (16), with P-Recoverability ranked above Economy of Pronunciation. P-Recoverability protects copy material that is independently associated with PF content, whereas Economy of Pronunciation deletes copy material that is not so protected.

- (16) a. P-Recoverability (PR)
 Copy material associated with independent PF content must be pronounced.
 b. Economy of Pronunciation (EP)
 Copy material not associated with independent PF content is deleted.

The ranking is given in (17). PR is a recoverability requirement, while EP is an elsewhere deletion principle. If the two principles had equal status, the grammar would not distinguish between PF-required copy material and redundant copy material. The relevant derivation therefore lets PR protect the material needed for the surface VD form, and lets EP delete the remaining material.

- (17)
- PR $\gg$ EP

For SpXP VD, the PF computation is also constrained by the STRAY AFFIX FILTER (Lasnik 1995). The perfective marker *-le* cannot remain without a morphophonological host. Since the surface host is not the bare verb *san*, but the complex lower M-word *san-qu*, the derivation must first create the lower verbal-spatial M-word and only then attach *-le*. This motivates the ordered PF sequence in (18).

- (18)
- Head Movement $\gg$ Chain Reduction / copy partition
 $\gg$ Preorder + Ext $\gg$ Local Dislocation

The input to PF is the copy chain in (19). Narrow syntax has created a phrasal movement dependency: the predicate-sized constituent moves to the low post-subject position, leaving a lower copy in the verbal domain.

- (19)
- $$\left[\begin{array}{l} \text{TP Piet [FP VP}_H[\text{san bu [SpXP}_{\text{DIR}} \text{qu gongyuan}]] \\ \text{[AspP -le [vP \langle DP}_{\text{SUBJ}} \rangle [v' v^0 \text{VP}_L[\text{san bu [SpXP}_{\text{DIR}} \text{qu gongyuan}]]]]]] \end{array} \right]$$

The first PF operation is Head Movement of the lower verbal head to v^0 (Boeckx and Stjepanovic 2001). This step is necessary because the lower verbal occurrence must survive Chain Reduction in order to provide part of the eventual host for *-le*.

(20)

$$\begin{aligned} \text{Input: } & [v^0 v^0 [\text{VP}_L \text{ san bu } [\text{SpXP}_{\text{DIR}} \text{ qu gongyuan}]]] \\ \text{Output: } & [v^0 \text{ san} + v^0] [\text{VP}_L \langle \text{san} \rangle \text{ bu } [\text{SpXP}_{\text{DIR}} \text{ qu gongyuan}]] \end{aligned}$$

After Head Movement, Chain Reduction applies. Since PR outranks EP, the post-subject higher copy is pronounced because it satisfies the PF requirement of the landing site ⁶. The lower verbal head is also protected, since it is needed for the aspectual host. EP deletes the remaining redundant copy material. In the higher copy, the directional SpXP is silenced. In the lower copy, the object *bu* and the VP-internal trace/copy of *san* are silenced. The lower SpXP remains pronounced in the base position of the chain.

(21)

$$\begin{aligned} \text{higher copy} & : \text{VP}_H = [\text{san bu } \langle \text{SpXP}_{\text{DIR}} \text{ qu gongyuan} \rangle] \\ \text{lower verbal head} & : v^0 = [\text{san}] \\ \text{lower copy} & : \text{VP}_L = [\langle \text{san} \rangle \langle \text{bu} \rangle [\text{SpXP}_{\text{DIR}} \text{ qu gongyuan}]] \end{aligned}$$

Given the output in (21), PF read-out follows Chain Reduction. Kuralian preorder realises Ext(*A*) before *A*'s ordered daughters, while Ext recursively returns the exponent of the associated head (Kural 2005). The higher copy thus yields *san-bu*. In the lower domain, deletion of *bu* makes head-moved *san* adjacent to *qu*, read out as $[_M \text{san-qu}]$. The string $[_M \text{le}] * [_M \text{san-qu}] * \text{gongyuan}$ then feeds Local Dislocation. Chain Reduction must precede read-out; otherwise, *bu* intervenes and the higher-copy SpXP is also linearised.

The final step is Local Dislocation. Local Dislocation is a post-linearisation PF operation that rebrackets two linearly adjacent morphological objects (Embick and Noyer 2001). I assume the typed version in (22).

(22)

$$X_\tau * Y_\tau \implies [Y_\tau - X_\tau] \quad \tau \in \{\text{M-Word, Subword}\}$$

For (15), the relevant input to Local Dislocation is the adjacency between the aspectual M-word $[_M \text{le}]$ and the lower verbal-spatial M-word $[_M \text{san-qu}]$. Local Dislocation attaches *-le* to the whole adjacent M-word, not to an arbitrary internal subpart of it.

(23)

$$[_M \text{le}] * [_M \text{san-qu}] \implies [_M \text{san-qu-le}]$$

4. Conclusion

This paper has developed a Distributed Morphology-compatible analysis of Mandarin verb doubling in which phrasal movement feeds an ordered sequence of PF operations. Locative and directional VD share the VP-fronting source of other VD patterns, but their surface realisation depends on the predicate-internal licensing of the lower SpXP and on copy partition, preorder read-out, and Local Dislocation. The analysis therefore derives the attested doubling strings while constraining the co-occurrence of spatial arguments with other postverbal material. Co-occurrence is possible when a spatial argument and an adjunct occupy distinct structural positions: as Yan (2025, 64, ex. 106) shows, a directional argument can combine with duration or

⁶I assume that clause-internal Topic/Focus-related PF requirements may be active. Following Hsu, I take FP to denote a clause-internal information-structural projection between TP and vP (Hsu 2013, p. 77).

frequency adjuncts. Its incompatibility with a manner adjunct, by contrast, potentially follows from an independent requirement associated with *de*. More generally, the proposal links the event-structural licensing of SpXPs in syntax to their selective realisation across copies at PF.

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